US012403272B2

(12) **United States Patent**
Alshaiba Saleh Ghannam Almazrouei et al.

(10) **Patent No.: US 12,403,272 B2**
(45) **Date of Patent: *Sep. 2, 2025**

(54) **ULTRASONIC MIST INHALER**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

This patent is subject to a terminal disclaimer.

(21) Appl. No.: **18/537,694**

(22) Filed: **Dec. 12, 2023**

(65) **Prior Publication Data**

US 2024/0100269 A1 Mar. 28, 2024

Related U.S. Application Data

(63) Continuation of application No. 16/959,958, filed as application No. PCT/IB2019/060808 on Dec. 15, 2019, now Pat. No. 11,911,559.

(51) **Int. Cl.**
A61M 15/00 (2006.01)
A24F 40/05 (2020.01)
(Continued)

(52) **U.S. Cl.**
CPC ***A61M 15/0085*** (2013.01); ***A24F 40/05*** (2020.01); ***A24F 40/42*** (2020.01);
(Continued)

(58) **Field of Classification Search**
CPC A61M 15/0085; A61M 15/06; A61M 15/0021; A61M 2205/3331;
(Continued)

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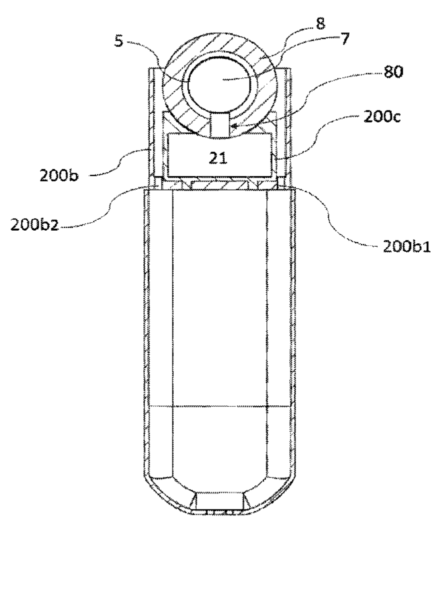
(57) **ABSTRACT**

An ultrasonic mist inhaler, including a liquid reservoir structure including a liquid chamber adapted to receive liquid to be atomized, the liquid chamber including a top surface representing a maximum level of the liquid chamber and a bottom surface representing a minimum level of the liquid chamber, a sonication chamber in fluid communication with the liquid chamber, the sonication chamber includes a piezoelectric transducer and a capillary material extending between the sonication chamber and the liquid chamber, wherein the piezoelectric transducer and the liquid chamber are arranged according to one of the following configurations: the piezoelectric transducer is disposed close to or at the bottom surface of the liquid chamber, the piezoelectric transducer is disposed close to or at the top surface of the liquid chamber as depicted in FIG. **3**.

7 Claims, 6 Drawing Sheets



(51) **Int. Cl.**

A24F 40/42	(2020.01)
A61M 11/00	(2006.01)
A61M 15/06	(2006.01)
A24F 40/10	(2020.01)
A24F 40/40	(2020.01)

(52) **U.S. Cl.**
CPC ***A61M 11/005*** (2013.01); ***A61M 15/06*** (2013.01); *A24F 40/10* (2020.01); *A24F 40/40* (2020.01); *A61M 15/0021* (2014.02); *A61M 2205/3331* (2013.01); *A61M 2205/50* (2013.01); *A61M 2205/8206* (2013.01)

(58) **Field of Classification Search**
CPC A61M 2205/50; A61M 2205/8206; A61M 2016/0018; A61M 11/005; A24F 40/05; A24F 40/10; A24F 40/40; A24F 40/42; A24B 15/167; B06B 1/20
See application file for complete search history.

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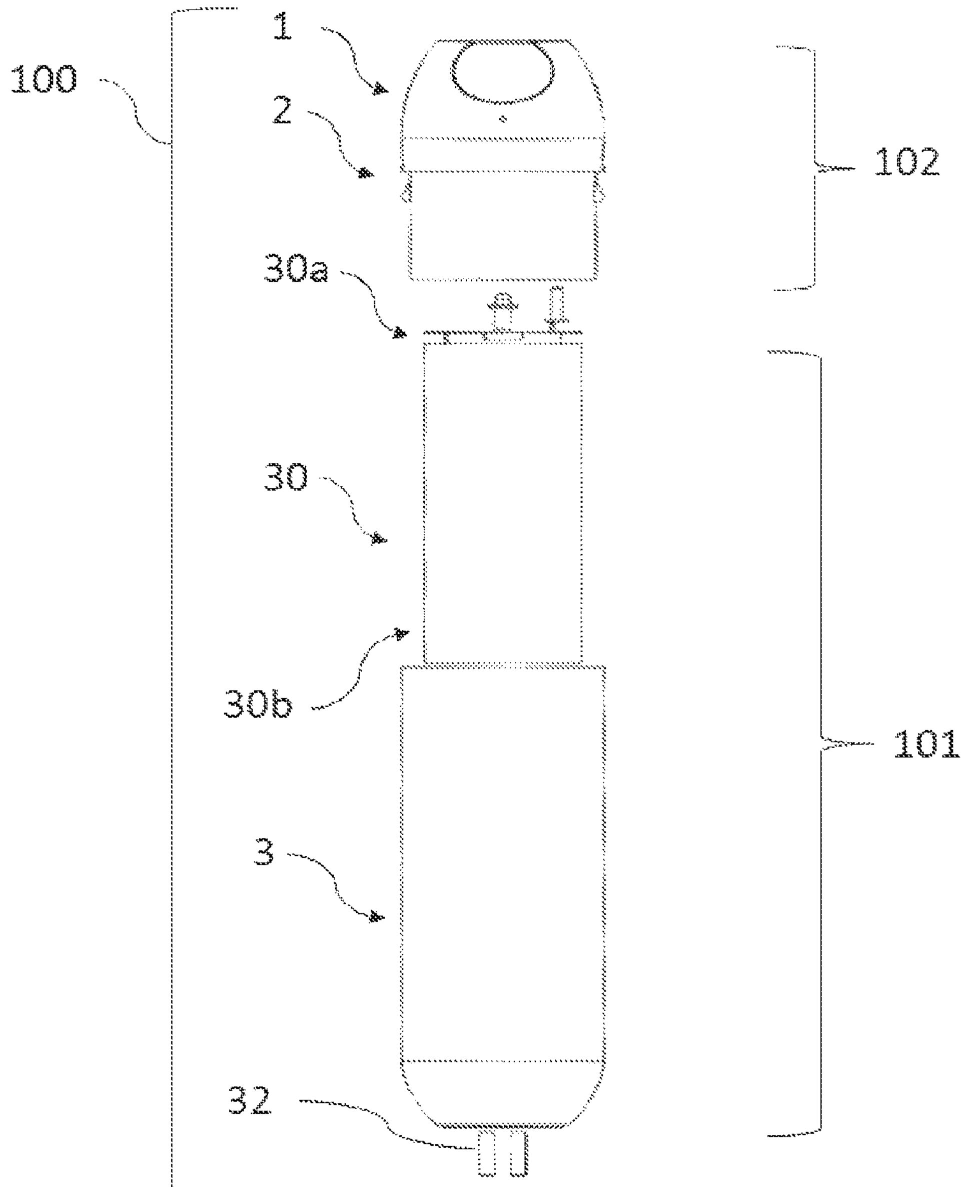

FIG. 1

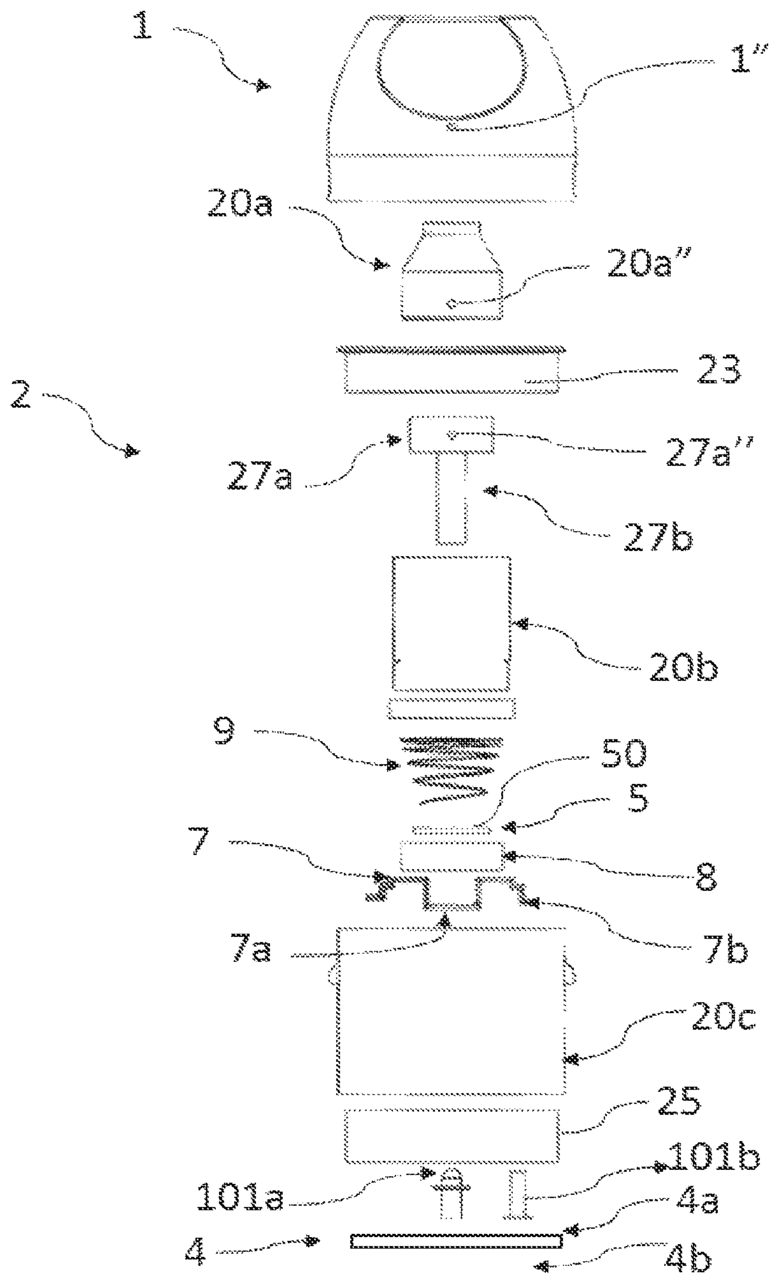
1
1″
20a
20a″
23
2
27a″
27a
27b
20b
9
50
5
7
8
7a
7b
20c
25
101b
101a
4a
4
4b

FIG. 2

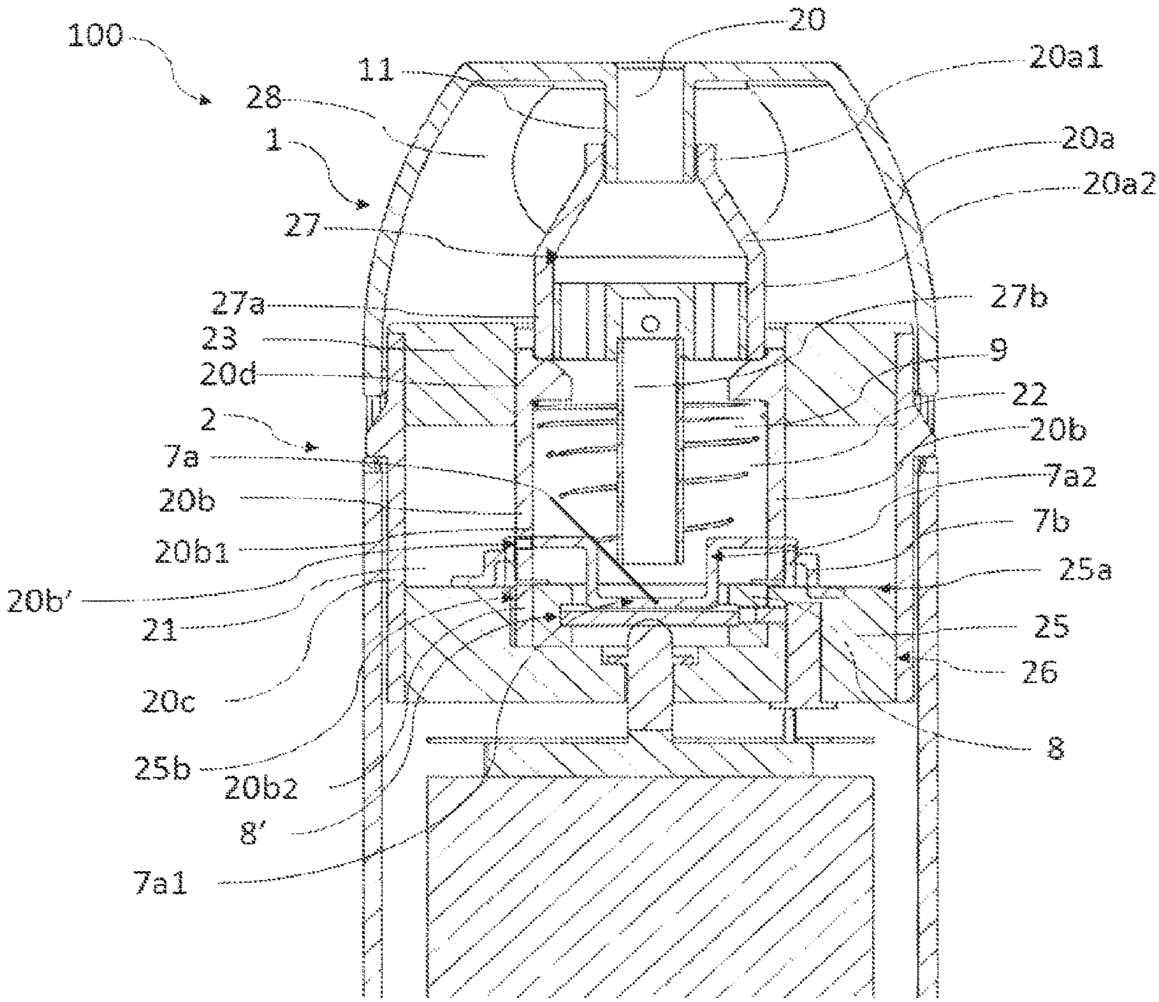
100
20
11
20a1
28
1
20a
20a2
27
27a
27b
23
20d
9
2
22
20b
7a
7a2
20b
7b
20b1
25a
20b'
21
25
26
20c
8
25b
20b2
8'
7a1

FIG. 3

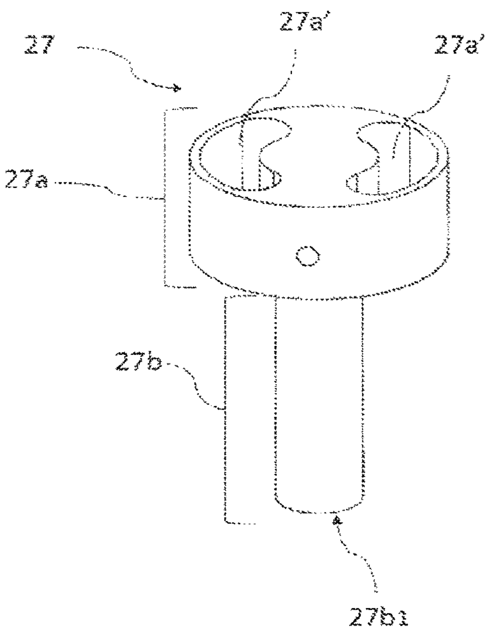

FIG. 4A

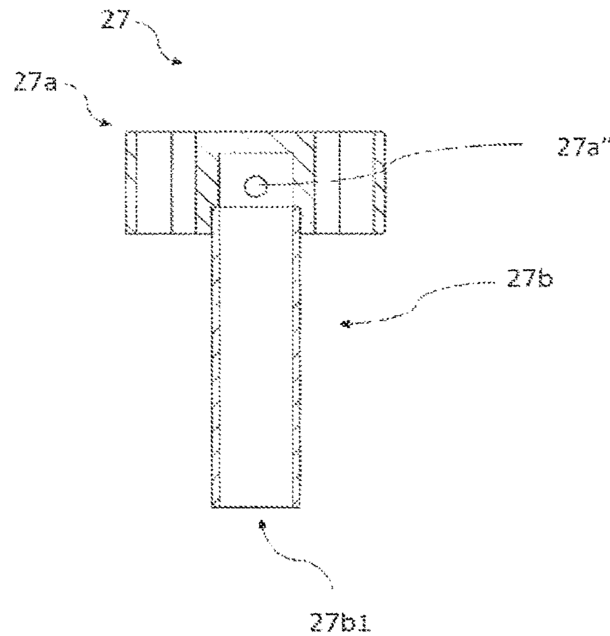

FIG. 4B

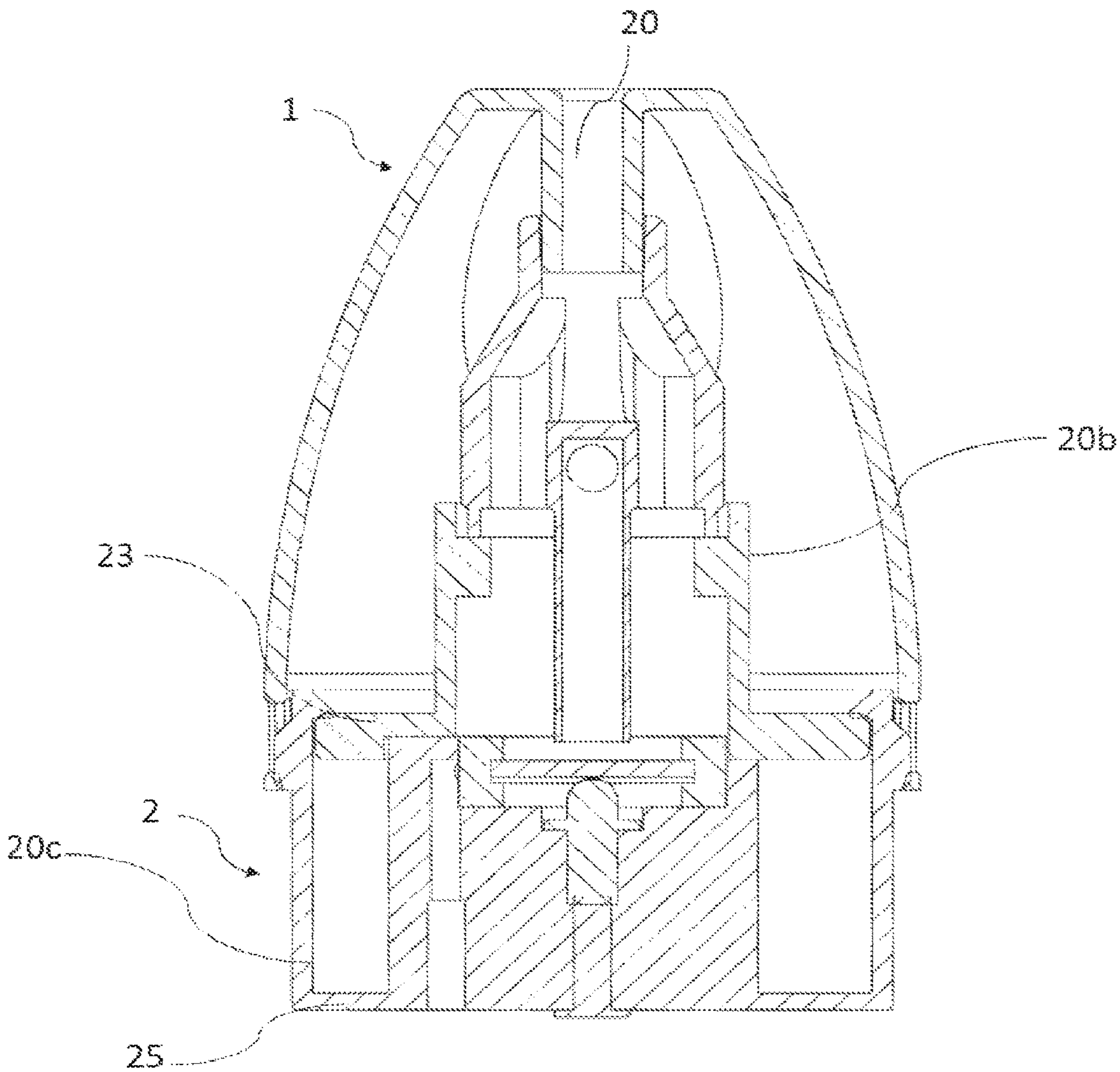

FIG. 5

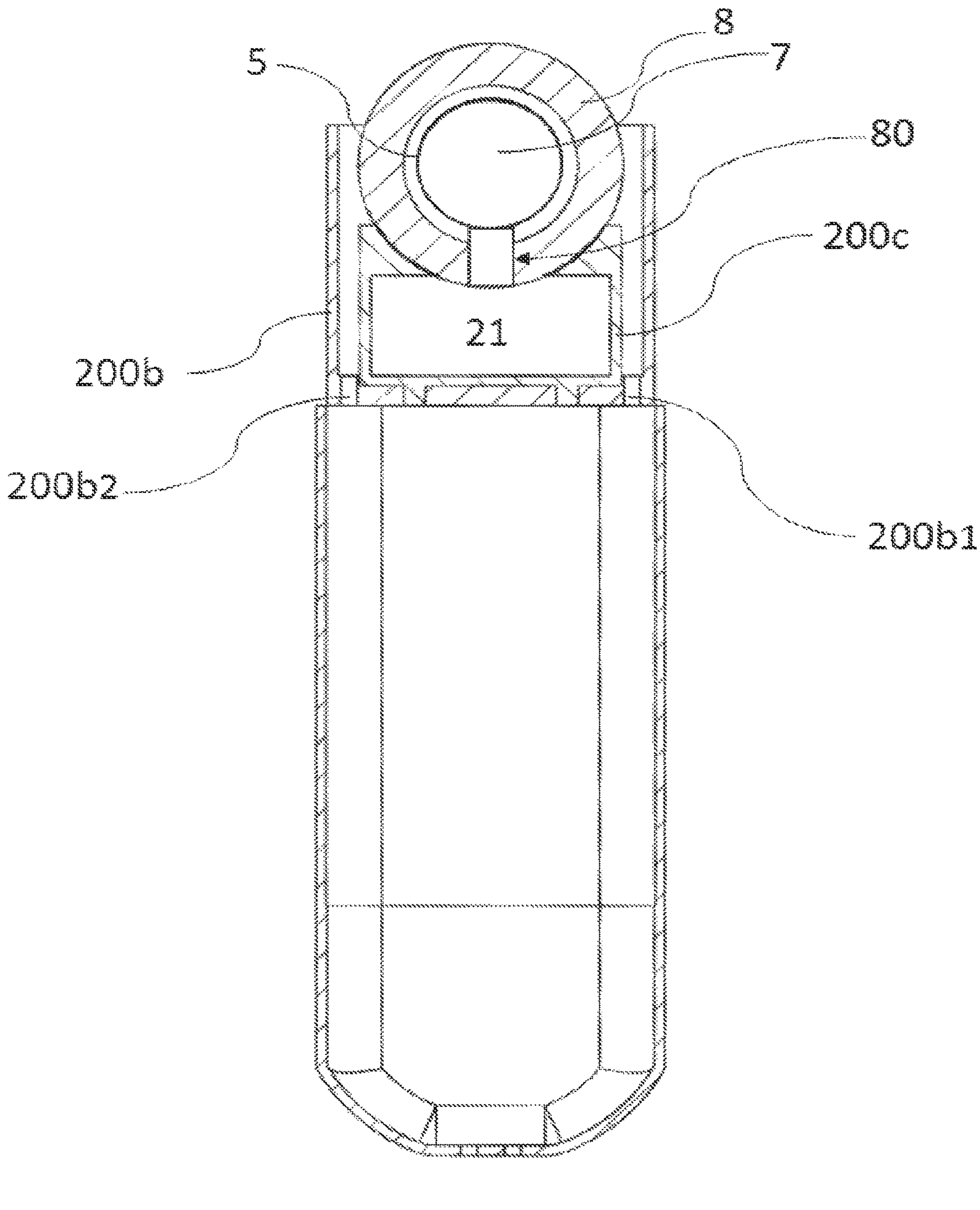

FIG. 6

ULTRASONIC MIST INHALER

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of application Ser. No. 16/959,958, filed Jul. 2, 2020; which is a National Phase Application under 35 U.S.C. § 371 of International Application No. PCT/IB2019/060808, filed Dec. 15, 2019, the entire contents of which are incorporated herein by reference.

FIELD OF THE DISCLOSED TECHNOLOGY

The invention relates to an ultrasonic mist inhaler for atomizing a liquid by ultrasonic vibrations.

BACKGROUND

Electronic vaporizing inhalers are becoming popular among smokers who also want to avoid the tar and other harsh chemicals associated with traditional cigarettes and who wish to satisfy the craving for nicotine. Electronic vaporizing inhalers may contain liquid nicotine, which is typically a mixture of nicotine oil, a solvent, water, and sometimes flavoring. When the user draws, or inhales, on the electronic vaporizing inhaler, the liquid nicotine is drawn into a vaporizer where it is heated into a vapor. As the user draws on the electronic vaporizing inhaler, the vapor containing the nicotine is inhaled. Such electronic vaporizing inhalers may have medical purpose.

Electronic vaporizing inhalers and other vapor inhalers typically have similar designs. Most electronic vaporizing inhalers feature a liquid nicotine reservoir with an interior membrane, such as a capillary element, typically cotton, that holds the liquid nicotine so as to prevent leaking from the reservoir. Nevertheless, these cigarettes are still prone to leaking because there is no obstacle to prevent the liquid from flowing out of the membrane and into the mouthpiece. A leaking electronic vaporizing inhaler is problematic for several reasons. As a first disadvantage, the liquid can leak into the electronic components, which can cause serious damage to the device. As a second disadvantage, the liquid can leak into the electronic vaporizing inhaler mouthpiece, and the user may inhale the unvaporized liquid.

Electronic vaporizing inhalers are also known for providing inconsistent doses between draws. The aforementioned leaking is one cause of inconsistent doses because the membrane may be oversaturated or undersaturated near the vaporizer. If the membrane is oversaturated, then the user may experience a stronger than desired dose of vapor, and if the membrane is undersaturated, then the user may experience a weaker than desired dose of vapor. Additionally, small changes in the strength of the user's draw may provide stronger or weaker doses. Inconsistent dosing, along with leaking, can lead to faster consumption of the vaping liquid.

Additionally, conventional electronic vaporizing inhalers tend to rely on inducing high temperatures of a metal heating component configured to heat a liquid in the e-cigarette, thus vaporizing the liquid that can be breathed in. Problems with conventional electronic vaporizing inhalers may include the possibility of burning metal and subsequent breathing in of the metal along with the burnt liquid. In addition, some may not prefer the burnt smell caused by the heated liquid.

Electronic vaporizing inhalers are generally designed so that the liquid nicotine reservoir is arranged away from the metal heating component to prevent heating the unused liquid in the reservoir. This arrangement makes the inhaler device cumbersome and more complex to produce.

In the electronic vaporizing inhalers, the vapor chamber is generally distant from the liquid reservoir and the mouthpiece to prevent heating the liquid or the vapor before the user drawn.

Thus, a need exists in the art for an electronic vaporizing inhaler that is better able to withstand these disadvantages.

BRIEF SUMMARY

According to one aspect of the invention, an ultrasonic mist inhaler, comprises:

- a liquid reservoir structure comprising a liquid chamber adapted to receive liquid to be atomized,
- the liquid chamber comprising a top surface representing the maximum level of the liquid chamber and the bottom surface representing the minimum level of the liquid chamber,
- a sonication chamber in fluid communication with the liquid chamber,
- the sonication chamber comprising means of ultrasonic vibrations and a capillary element extending between the sonication chamber and the liquid chamber,
- wherein the means of ultrasonic vibrations and the liquid chamber are arranged according to one of the following configurations:
  - the means of ultrasonic vibrations are disposed close to or at the bottom surface of the liquid chamber,
  - the means of ultrasonic vibrations are disposed close to or at the top surface of the liquid chamber.

The arrangement of the means of ultrasonic vibrations disposed close to or at the bottom surface of the liquid chamber renders the fluid passage to the means of ultrasonic vibrations by gravity faster.

The arrangement of the means of ultrasonic vibrations disposed close to or at the top surface of the liquid chamber renders the mist passage to a mouthpiece shorter.

The expression "means of ultrasonic vibrations" is similar to the expression "ultrasonic oscillation component" used in the patent application PCT/IB2019/055192.

In the ultrasonic mist inhaler, in the arrangement close to or at the top surface, the means of ultrasonic vibrations may have at least an atomization surface parallel to the top surface.

In the ultrasonic mist inhaler, in the arrangement close to or at the top surface, the means of ultrasonic vibrations may have at least an atomization surface perpendicular to the top surface.

The atomization surface perpendicular to the top surface permits smaller thickness of the inhaler.

In the configuration of the atomization surface perpendicular to the top surface, the inhaler may comprise one of the following features taken alone or in combination:

- the liquid reservoir structure may comprise an inner container and an outer container wherein the inner container is surrounded by the outer container, the inner container forms the liquid reservoir,
- the capillary element is wrapped around the means of ultrasonic vibrations,
- the means of ultrasonic vibrations is supported by an elastic member,
- the elastic member is formed from an annular plate-shaped rubber,
- the elastic member has an inner hole wherein a groove is designed for maintaining the means of ultrasonic vibrations,

the inner container has an opening into which the elastic member is introduced at least partly, the elastic member having a radial groove wherein the capillary element passes and penetrates the liquid chamber,

the elastic member is inserted into the opening of the inner container by tight adjustment.

It is noted that the expression "mist" used in the invention means the liquid is not heated as usually in traditional inhalers known from the prior art. In fact, traditional inhalers use heating elements to heat the liquid above its boiling temperature to produce a vapor, which is different from a mist.

In fact, when sonicating liquids at high intensities, the sound waves that propagate into the liquid media result in alternating high-pressure (compression) and low-pressure (rarefaction) cycles, at different rates depending on the frequency. During the low-pressure cycle, high-intensity ultrasonic waves create small vacuum bubbles or voids in the liquid. When the bubbles attain a volume at which they can no longer absorb energy, they collapse violently during a high-pressure cycle. This phenomenon is termed cavitation. During the implosion very high pressures are reached locally. At cavitation, broken capillary waves are generated, and tiny droplets break the surface tension of the liquid and are quickly released into the air, taking mist form.

The ultrasonic mist inhaler according to the invention, wherein said liquid to be received in the liquid chamber comprises 57-70% (w/w) vegetable glycerin and 30-43% (w/w) propylene glycol, said propylene glycol including nicotine and flavorings.

An ultrasonic mist inhaler or a personal ultrasonic atomizer device, comprising:

a liquid reservoir structure comprising a liquid chamber or cartridge adapted to receive liquid to be atomized,

a sonication chamber in fluid communication with the liquid chamber or cartridge,

wherein said liquid to be received in the liquid chamber comprises 57-70% (w/w) vegetable glycerin and 30-43% (w/w) propylene glycol, said propylene glycol including nicotine and flavorings.

BRIEF DESCRIPTION OF THE DRAWINGS

Some embodiments are illustrated by way of example and not limitation in the figures of the accompanying drawings.

FIG. **1** is an exploded view of components of the ultrasonic mist inhaler according to a first arrangement of means of ultrasonic vibrations disposed close to or at a bottom surface of a liquid chamber.

FIG. **2** is an exploded view of components of the inhaler liquid reservoir structure according to the first arrangement.

FIG. **3** is a cross section view of components of the inhaler liquid reservoir structure according to FIG. **1** or **2** wherein means of ultrasonic vibrations are arranged according to a first embodiment.

FIG. **4**A is an isometric view of an airflow member of the inhaler liquid reservoir structure according to FIGS. **2** and **3**.

FIG. **4**B is a cross section view of the airflow member shown in FIG. **4**A.

FIG. **5** is a cross section view of a second arrangement of means of ultrasonic vibrations disposed close to or at a top surface of the liquid chamber.

FIG. **6** is a cross section view of a second arrangement of means of ultrasonic vibrations disposed close to or at the top surface of the liquid chamber.

DETAILED DESCRIPTION

The foregoing summary, as well as the following detailed description of certain embodiments of the present invention, will be better understood when read in conjunction with the appended drawings.

As used herein, an element recited in the singular and preceded with the word "a" or "an" should be understood as not excluding plural of said elements, unless such exclusion is explicitly stated. Furthermore, the references to "one embodiment" of the present invention are not intended to be interpreted as excluding the existence of additional embodiments that also incorporate the recited features. Moreover, unless explicitly stated to the contrary, embodiments "comprising" or "having" an element or a plurality of elements having a particular property may include additional such elements not having that property.

The present invention is directed to an ultrasonic mist inhaler. The description of the invention and accompanying figures will be directed to the electronic vaporizing inhaler embodiment; however, other embodiments are envisioned, such as an inhaler for hookah, flavored liquids, medicine, and herbal supplements. Additionally, the device can be packaged to look like an object other than a cigarette. For instance, the device could resemble another smoking instrument, such as a pipe, water pipe, or slide, or the device could resemble another non-smoking related object.

Ultrasonic mist inhalers are either disposable or reusable. The term "reusable" as used herein implies that the energy storage device is rechargeable or replaceable or that the liquid is able to be replenished either through refilling or through replacement of the liquid reservoir structure. Alternatively, in some embodiments reusable electronic device is both rechargeable and the liquid can be replenished. A disposable embodiment will be described first, followed by a description of a reusable embodiment.

Conventional electronic vaporizing inhaler tend to rely on inducing high temperatures of a metal component configured to heat a liquid in the inhaler, thus vaporizing the liquid that can be breathed in. The liquid typically contains nicotine and flavorings blended into a solution of propylene glycol (PG) and vegetable glycerin (VG), which is vaporized via a heating component at high temperatures. Problems with conventional inhaler may include the possibility of burning metal and subsequent breathing in of the metal along with the burnt liquid. In addition, some may not prefer the burnt smell or taste caused by the heated liquid.

In contrast, aspects of the present disclosure include an ultrasonic mist inhaler that atomizes the liquid through ultrasonic vibrations, which produces micro water bubbles in the liquid. When the bubbles come into contact with ambient air, water droplets of about 0.25 to 0.5 microns spray into the air, thereby generating micro-droplets that can be absorbed through breathing, similar to breathing in a mist.

No heating elements are involved, thereby leading to no burnt elements and reducing second-hand smoke effects.

FIG. **1** to FIG. **4** illustrate a first arrangement of means of ultrasonic vibrations disposed close to or at a bottom surface of a liquid chamber.

FIG. **1** depicts a disposable ultrasonic mist inhaler embodiment **100** of the invention. As can be seen in FIG. **1**, the ultrasonic mist inhaler **100** has a cylindrical body with a relatively long length as compared to the diameter. In terms of shape and appearance, the ultrasonic mist inhaler **100** is designed to mimic the look of a typical cigarette. For instance, the inhaler can feature a first portion **101** that primarily simulates the tobacco rod portion of a cigarette and a second portion **102** that primarily simulates a filter. In the disposable embodiment of the invented device, the first portion and second portion are regions of a single, but-separable device. The designation of a first portion **101** and a second portion **102** is used to conveniently differentiate the components that are primarily contained in each portion.

As can be seen in FIG. **1**, the ultrasonic mist inhaler comprises a mouthpiece **1**, a liquid reservoir structure **2** and a casing **3**. The first portion **101** comprises the casing **3** and the second portion **102** comprises the mouthpiece **1** and the reservoir structure **2**.

The first portion **101** contains the power supply energy.

An electrical storage device **30** powers the ultrasonic mist inhaler **100**. The electrical storage device **30** can be a battery, including but not limited to a lithium-ion, alkaline, zinc-carbon, nickel-metal hydride, or nickel-cadmium battery; a super capacitor; or a combination thereof. In the disposable embodiment, the electrical storage device **30** is not rechargeable, but, in the reusable embodiment, the electrical storage device **30** would be selected for its ability to recharge. In the disposable embodiment, the electrical storage device **30** is primarily selected to deliver a constant voltage over the life of the inhaler **100**. Otherwise, the performance of the inhaler would degrade over time. Preferred electrical storage devices that are able to provide a consistent voltage output over the life of the device include lithium-ion and lithium polymer batteries.

The electrical storage device **30** has a first end **30***a* that generally corresponds to a positive terminal and a second end **30***b* that generally corresponds to a negative terminal. The negative terminal is extending to the first end **30***a*.

Because the electrical storage device **30** is located in the first portion **101** and the liquid reservoir structure **2** is located in the second portion **102**, the joint needs to provide electrical communication between those components. In the present invention, electrical communication is established using at least an electrode or probe that is compressed together when the first portion **101** is tightened into the second portion **102**.

In order for this embodiment to be reusable, the electrical storage device **30** is rechargeable. The casing **3** contains a charging port **32**.

The integrated circuit **4** has a proximal end **4***a* and a distal end **4***b*. The positive terminal at the first end **30***a* of the electrical storage device **30** is in electrical communication with a positive lead of the flexible integrated circuit **4**. The negative terminal at the second end **30***b* of the electrical storage device **30** is in electrical communication with a negative lead of the integrated circuit **4**. The distal end **4***b* of the integrated circuit **4** comprise a microprocessor. The microprocessor is configured to process data from a sensor, to control a light, to direct current flow to means of ultrasonic vibrations **5** in the second portion **102**, and to terminate current flow after a preprogrammed amount of time.

The sensor detects when the ultrasonic mist inhaler **100** is in use (when the user draws on the inhaler) and activates the microprocessor. The sensor can be selected to detect changes in pressure, air flow, or vibration. In a preferred embodiment, the sensor is a pressure sensor. In the digital embodiment, the sensor takes continuous readings which in turn requires the digital sensor to continuously draw current, but the amount is small and overall battery life would be negligibly affected.

Additionally, the integrated circuit **4** may comprise a H bridge, preferably formed by 4 MOSFETs to convert a direct current into an alternate current at high frequency.

Referring to FIG. **2** and FIG. **3**, illustrations of a liquid reservoir structure **2** according to an embodiment are shown. The liquid reservoir structure **2** comprises a liquid chamber **21** adapted to receive liquid to be atomized and a sonication chamber **22** in fluid communication with the liquid chamber **21**.

In the embodiment shown, the liquid reservoir structure **2** comprises an inhalation channel **20** providing an air passage from the sonication chamber **22** toward the surroundings.

As an example of sensor position, the sensor may be located in the sonication chamber **22**.

The inhalation channel **20** has a frustoconical element **20***a* and an inner container **20***b*.

As depicted in FIGS. **4**A and **4**B, further the inhalation channel **20** has an airflow member **27** for providing air flow from the surroundings to the sonication chamber **22**.

The airflow member **27** has an airflow bridge **27***a* and an airflow duct **27***b* made in one piece, the airflow bridge **27***a* having two airway openings **27***a*' forming a portion of the inhalation channel **20** and the airflow duct **27***b* extending in the sonication chamber **22** from the airflow bridge **27***a* for providing the air flow from the surroundings to the sonication chamber.

The airflow bridge **27***a* cooperates with the frustoconical element **20***a* at the second diameter **20***a***2**.

The airflow bridge **27***a* has two opposite peripheral openings **27***a*" providing air flow to the airflow duct **27***b*.

The cooperation with the airflow bridge **27***a* and the frustoconical element **20***a* is arranged so that the two opposite peripheral openings **27***a*" cooperate with complementary openings **20***a*" in the frustoconical element **20***a*.

The mouthpiece **1** and the frustoconical element **20***a* are radially spaced and an airflow chamber **28** is arranged between them.

As depicted in FIGS. **1** and **2**, the mouthpiece **1** has two opposite peripheral openings **1**".

The peripheral openings **27***a*", **20***a*", **1**" of the airflow bridge **27***a*, the frustoconical element **20***a* and the mouthpiece **1** directly supply maximum air flow to the sonication chamber **22**.

The frustoconical element **20***a* includes an internal passage, aligned in the similar direction as the inhalation channel **20**, having a first diameter **20***a***1** less than that of a second diameter **20***a***2**, such that the internal passage reduces in diameter over the frustoconical element **20***a*.

The frustoconical element **20***a* is positioned in alignment with the means of ultrasonic vibrations **5** and a capillary element **7**, wherein the first diameter **20***a***1** is linked to an inner duct **11** of the mouthpiece **1** and the second diameter **20***a***2** is linked to the inner container **20***b*.

The inner container **20***b* has an inner wall delimiting the sonication chamber **22** and the liquid chamber **21**.

The liquid reservoir structure **2** has an outer container **20***c* delimiting the outer wall of the liquid chamber **21**.

The inner container **20***b* and the outer container **20***c* are respectively the inner wall and the outer wall of the liquid chamber **21**.

The liquid reservoir structure **2** is arranged between the mouthpiece **1** and the casing **3** and is detachable from the mouthpiece **1** and the casing **3**.

The liquid reservoir structure **2** and the mouthpiece **1** or the casing **3** may include complimentary arrangements for engaging with one another; further such complimentary arrangements may include one of the following: a bayonet type arrangement; a threaded engaged type arrangement; a magnetic arrangement; or a friction fit arrangement; wherein the liquid reservoir structure **2** includes a portion of the arrangement and the mouthpiece **1** or the casing **3** includes the complimentary portion of the arrangement.

In the reusable embodiment, the components are substantially the same. The differences in the reusable embodiment vis-a-vis the disposable embodiment are the accommodations made to replace the liquid reservoir structure **2**.

As shown in FIG. **3**, the liquid chamber **21** has a top wall **23** and a bottom wall **25** closing the inner container **20***b* and the outer container **20***c* of the liquid chamber **21**.

The capillary element **7** is arranged between a first section **20***b***1** and a second section **20***b***2** of the inner container **20***b*.

The capillary element **7** has a flat shape extending from the sonication chamber to the liquid chamber.

As depicted in FIG. **2** or **3**, the capillary element **7** comprises a central portion **7***a* in U-shape and a peripheral portion **7***b* in L-shape.

The L-shape portion **7***b* extends into the liquid chamber **21** on the inner container **20***b* and along the bottom wall **25**.

The U-shape portion **7***a* is contained into the sonication chamber **21**. The U-shape portion **7***a* on the inner container **20***b* and along the bottom wall **25**.

In the ultrasonic mist inhaler, the U-shape portion **7***a* has an inner portion **7***a***1** and an outer portion **7***a***2**, the inner portion **7***a***1** being in surface contact with an atomization surface **50** of the means of ultrasonic vibrations **5** and the outer portion **7***a***2** being not in surface contact with the means of ultrasonic vibrations **5**.

The bottom wall **25** of the liquid chamber **21** is a bottom plate **25** closing the liquid chamber **21** and the sonication chamber **22**. The bottom plate **25** is sealed, thus preventing leakage of liquid from the sonication chamber **22** to the casing **3**.

The bottom plate **25** has an upper surface **25***a* having a recess **25***b* on which is inserted an elastic member **8**. The means of ultrasonic vibrations **5** are supported by the elastic member **8**. The elastic member **8** is formed from an annular plate-shaped rubber having an inner hole **8'** wherein a groove is designed for maintaining the means of ultrasonic vibrations **5**.

The top wall **23** of the liquid chamber **21** is a cap **23** closing the liquid chamber **23**.

The top wall **23** has a top surface **23** representing the maximum level of the liquid that the liquid chamber **21** may contain and the bottom surface **25** representing the minimum level of the liquid in the liquid chamber **21**.

The top wall **23** is sealed, thus preventing leakage of liquid from the liquid chamber **21** to the mouthpiece **1**.

The top wall **23** and the bottom wall **25** are fixed to the liquid reservoir structure **2** by means of fixation such as screws, glue, or friction.

As depicted in FIG. **3**, the elastic member **8** is in line contact with the means of ultrasonic vibrations **5** and prevents contact between the means of ultrasonic vibrations **5** and the inhaler walls, suppression of vibrations of the liquid reservoir structure are more effectively prevented. Thus, fine particles of the liquid atomized by the atomizing member can be sprayed farther.

As depicted in FIG. **3**, the inner container **20***b* has openings **20***b*' between the first section **20***b***1** and the second section **20***b***2** from which the capillary element **7** is extending from the sonication chamber **21**. The capillary element **7** absorbs liquid from the liquid chamber **21** through the apertures **20***b*'. The capillary element **7** is a wick. The capillary element **7** transports liquid to the sonication chamber **22** via capillary action. Preferably the capillary element **7** is made of bamboo fibers; however, cotton, paper, or other fiber strands could be used for a wick material.

In one embodiment of the ultrasonic mist inhaler **100**, as can be seen in FIG. **3**, the means of ultrasonic vibrations **5** are disposed directly below the capillary element **7**.

The means of ultrasonic vibrations **5** may be a transducer. For example, the means of ultrasonic vibrations **5** may be a piezoelectric transducer, preferably designed in a circular plate-shape. The material of the piezoelectric transducer is preferably in ceramic.

A variety of transducer materials can also be used for the means of ultrasonic vibrations **5**.

The end of the airflow duct **27***b***1** faces the means of ultrasonic vibrations **5**. The means of ultrasonic vibrations **5** are in electrical communication with electrical contactors **101***a*, **101***b*. It is noted that, the distal end **4***b* of the integrated circuit **4** has an inner electrode and an outer electrode. The inner electrode contacts the first electrical contact **101***a* which is a spring contact probe, and the outer electrode contacts the second electrical contact **101***b* which is a side pin. Via the integrated circuit **4**, the first electrical contact **101***a* is in electrical communication with the positive terminal of the electrical storage device **30** by way of the microprocessor, while the second electrical contact **101***b* is in electrical communication with the negative terminal of the electrical storage device **30**.

The electrical contacts **101***a*, **101***b* crossed the bottom plate **25**. The bottom plate **25** is designed to be received inside the perimeter wall **26** of the liquid reservoir structure **2**. The bottom plate **25** rests on complementary ridges, thereby creating the liquid chamber **21** and sonication chamber **22**.

The inner container **20***b* comprises a circular inner slot **20***d* on which a mechanical spring is applied.

By pushing the central portion **7***a***1** onto the means of ultrasonic vibrations **5**, the mechanical spring **9** ensures a contact surface between them.

The liquid reservoir structure **2** and the bottom plate **25** can be made using a variety of thermoplastic materials.

When the user draws on the ultrasonic mist inhaler **100**, an air flow is drawn from the peripheral openings **1"** and penetrates the airflow chamber **28**, passes the peripheral openings **27***a*" of the airflow bridge **27***a* and the frustoconical element **20***a* and flows down into the sonication chamber **22** via the airflow duct **27***b* directly onto the capillary element **7**. At the same time, the liquid is drawn from the reservoir chamber **21** by capillarity, through the plurality of apertures **20***b*', and into the capillary element **7**. The capillary element **7** brings the liquid into contact with the means of ultrasonic vibrations **5** of the inhaler **100**. The user's draw also causes the pressure sensor to activate the integrated circuit **4**, which directs current to the means of ultrasonic vibrations **5**. Thus, when the user draws on the mouthpiece **1** of the inhaler **100**, two actions happen at the same time. Firstly, the sensor activates the integrated circuit **4**, which triggers the means of ultrasonic vibrations **5** to begin vibrating. Secondly, the draw reduces the pressure outside the reservoir chamber **21** such that flow of the liquid through the apertures **20***b*' begins, which saturates the capillary element **7**. The capillary element **7** transports the liquid to the means of ultrasonic vibrations **5**, which causes bubbles to form in a capillary channel by the means of ultrasonic vibrations **5** and mist the liquid. Then, the mist liquid is drawn by the user.

The ultrasonic mist inhaler **100** of the present disclosures is a more powerful version of current portable medical nebulizers, in the shape and size of current e-cigarettes and with a particular structure for effective vaporization. It is a healthier alternative to cigarettes and current e-cigarettes products.

The ultrasonic mist inhaler **100** of the present disclosures has particular applicability for those who use electronic inhalers as a means to quit smoking and reduce their nicotine dependency. The ultrasonic mist inhaler **100** provides a way to gradually taper the dose of nicotine.

FIG. **5** illustrate a second arrangement of means of ultrasonic vibrations disposed close to or at a top surface of the liquid chamber.

The mouthpiece **1** is extending down to the bottom end of the inner container **20***b*. While the liquid chamber **21** is delimited by the outer container **20***c* and an inner peripheral container **20***e* joined by a bottom wall **25**. The liquid chamber is closed by a top wall **23**.

The top wall **23** has the top surface **23** representing the maximum level of the liquid that the liquid chamber **21** may contain and the bottom surface **25** representing the minimum level of the liquid in the liquid chamber **21**. The top wall **23** is sealed, thus preventing leakage of liquid from the liquid chamber **21** to the mouthpiece **1**.

As depicted in FIG. **5**, the means of ultrasonic vibrations **5** has a flat shape having an atomization surface **50** parallel to the top surface **23** and disposed close to or at the top surface **23**.

The top wall **23** is fixed to the liquid reservoir structure **2** by means of fixation such as screws or glue.

Contrary to the first arrangement, the capillary element **7** is not shown to make the illustrated features more clear.

Of course, the capillary element **7** may have the same configuration that the first arrangement. In this second arrangement, the L-shape is extending down to the bottom surface **25**.

FIG. **6** illustrate a second arrangement of means of ultrasonic vibrations disposed close to or at a top surface of the liquid chamber.

In the second arrangement, the liquid reservoir structure **2** may comprise an inner container **200***c* and an outer container **200***b* wherein the inner container **200***c* is surrounded by the outer container **200***b*, the inner container **200***c* forms the liquid chamber **21**.

The capillary element **7** is wrapped around the circular shape of the means of ultrasonic vibrations **5**.

Further, the means of ultrasonic vibrations **5** is supported by an elastic member **8** formed from an annular plate-shaped rubber.

The elastic member **8** has an inner hole wherein a groove **80** is designed for maintaining the capillary element **7** and the means of ultrasonic vibrations **5**.

The inner container **200***c* has a flat opening into which the elastic member **8** is introduced at least partly, the elastic member **8** having a radial groove **80** wherein the capillary element **7** passes and penetrates the liquid chamber.

The inner container **200***c* is sealed with the elastic member **8**, for example, by glue or mastic disposed around the peripheral of the flat opening.

In this third arrangement, the means of electrical contacts rise through electrical contact holes **200***b***1** and **200***b***2** on either side of inner container **200***c* and connect with the rear of the means of ultrasonic vibrations **5**.

Other embodiments of the invented ultrasonic mist inhaler **100** are easily envisioned, including medicinal delivery devices.

It is to be understood that the above description is intended to be illustrative, and not restrictive. For example, the above-described embodiments may be used in combination with each other. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the invention without departing from its scope.

The invention claimed is:

1. An ultrasonic mist inhaler, comprising:

a liquid reservoir structure comprising a first container forming a liquid chamber adapted to receive liquid to be atomized, the first container comprises a top wall, a bottom wall, and a side wall positioned between the top wall and the bottom wall, the top wall comprises an interior surface, an exterior surface and an upward facing curved surface is formed between the exterior surface and the interior surface of the top wall, the upward facing curved surface forms an aperture to provide access to the liquid chamber, the liquid chamber is formed by the interior surface of the top wall, wherein the interior surface of the top wall is a top surface of the liquid chamber representing a maximum level of the liquid chamber, wherein the bottom wall comprises an interior surface that is a bottom surface of the liquid chamber representing a minimum level of the liquid chamber, and wherein the top surface being in a plane,

a sonication chamber in fluid communication with the liquid chamber, the sonication chamber comprising a piezoelectric transducer, wherein the piezoelectric transducer is disposed adjacent to the top surface of the liquid chamber, the piezoelectric transducer having an atomization surface in a plane perpendicular to the plane of the top surface of the liquid chamber,

a capillary material extending between the sonication chamber and the liquid chamber, and

an annular elastic support for supporting the piezoelectric transducer, the annular elastic support having a plate-shaped configuration and an annular wall surrounding at least a portion of the piezoelectric transducer, the annular elastic support having a passage extending radially through a portion of the annular wall,

wherein the aperture in the top wall of the first container is sized and configured to receive at least a lower wall portion of the annular wall therein, such that a first part of the lower wall portion of the annular wall is configured to engage the upward facing curved surface of the top wall of the liquid reservoir and a second part of the lower wall portion of the annular wall is positioned within the liquid chamber,

wherein a portion of the capillary material passes through the passage of the annular elastic support and penetrates the liquid chamber.

2. The ultrasonic mist inhaler according to claim **1**, wherein the first container is an inner container, and the liquid reservoir structure further comprises an outer container, and the inner container is surrounded by the outer container.

3. The ultrasonic mist inhaler according to claim **1**, wherein the capillary material is wrapped around the piezoelectric transducer.

4. The ultrasonic mist inhaler according to claim **1**, wherein the passage is formed by a groove that penetrates the lower wall portion of the annular wall of the annular elastic support.

5. The ultrasonic mist inhaler according to claim **4**, wherein the annular elastic support is inserted into the aperture by tight adjustment.

6. The ultrasonic mist inhaler according to claim **1**, wherein the annular elastic support having a radial groove, wherein the radial groove is configured to maintain the capillary material therein, the capillary material passing through the groove and penetrating the liquid chamber.

7. The ultrasonic mist inhaler according to claim **1**, wherein the liquid to be received in the liquid chamber comprises 57-70% (w/w) vegetable glycerin, 30-43% (w/w) propylene glycol, nicotine, and flavorings.

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